

Problem 1:

If X is a topological space and $Z \subset X$ is an irreducible closed subset prove that when $X = \operatorname{Spec}(R)$ for a ring R , every irreducible closed subset $Z \subset X$ has a unique generic point, given explicitly by $I(Z)$.

Proof.

Assume $X = \operatorname{Spec}(R)$ and Z is an irreducible closed subset. Shown in class since there is a bijection between prime ideals and irreducible closed subsets this implies that $Z = V(\mathfrak{p})$ for $\mathfrak{p} \in \operatorname{Spec}(R)$.

Consider $V(\mathfrak{p}) \cong \operatorname{Spec}(R/\mathfrak{p})$.

Since $\operatorname{Spec}(R/\mathfrak{p})$ is an integral domain this implies (0) is a generic point since (0) is prime in an integral domain. Consider $\overline{\{(0)\}} = V((0)) = \operatorname{Spec}(R/\mathfrak{p})$.

This is unique since for any $0 \neq \mathfrak{q} \in \operatorname{Spec}(R/\mathfrak{p}) \implies \overline{\{\mathfrak{q}\}} = V(\mathfrak{q})$ but since $\mathfrak{q} \neq 0 \implies \mathfrak{q} \not\subset (0)$ thus, $(0) \notin V(\mathfrak{q})$. So $\mathfrak{q}$ is not dense then going back up to $\operatorname{Spec}(R)$ we get that $\{\mathfrak{p}\}$ is the unique generic point. ■

Problem 2:**(a)**

Prove that $\mathfrak{p} \in \text{Spec}(R)$ is weakly associated prime of M if and only if there exists $m \in M$ s.t. $\mathfrak{p}$ is a generic point of an irreducible component of $\text{Supp}_R(m)$.

Proof.

Assume $\mathfrak{p} \in \text{Spec}(R)$ is a weakly associated prime of M . This implies that there exists an $m \in M$ s.t. $\mathfrak{p}$ is a minimal element of the set of prime ideals $\{\mathfrak{q} \in \text{Spec}(R) \mid \text{Ann}_R(m) \subset \mathfrak{q}\}$. Consider $\text{Supp}_R(m) = \text{Supp}_R(Rm) \implies \text{Supp}_R(Rm) = V(\text{Ann}_R(m))$ since Rm is finitely generated.

Now since $V(\text{Ann}_R(m)) \cong \text{Spec}(R/\text{Ann}_R(m))$, and $\mathfrak{p}$ is minimal over $\text{Ann}_R(m)$ this implies that $\overline{\mathfrak{p}}$ the image of $\mathfrak{p}$ in $\text{Spec}(R/\text{Ann}_R(m))$ is a minimal prime.

Proven in homework this implies that $\overline{\{\mathfrak{p}\}} = V(\overline{\mathfrak{p}})$ is an irreducible component of $\text{Spec}(R/\text{Ann}_R(m))$ which implies $V(\mathfrak{p})$ in $\text{Spec}(R)$ is an irreducible component of $V(\text{Ann}_R(m))$, so $\mathfrak{p}$ is a generic point of an irreducible component of $\text{Supp}_R(m)$.

Assume there exists $m \in M$ s.t. $\mathfrak{p}$ is a generic point of an irreducible component of $\text{Supp}_R(m)$. This implies that $V(\mathfrak{p})$ is an irreducible component of $\text{Supp}_R(m) = V(\text{Ann}_R(m))$. Which implies that $\mathfrak{p}$ is minimal over $V(\text{Ann}_R(m))$ by using the fact that $V(\text{Ann}_R(m)) \cong \text{Spec}(R/\text{Ann}_R(m))$ thus, $\mathfrak{p}$ is a weakly associated prime of M . ■

(b)

Prove the following conditions of $\mathfrak{p} \in \text{Spec}(R)$ are equivalent:

- i. $\mathfrak{p}$ is a weakly associated prime of M .
- ii. $\mathfrak{p}R_{\mathfrak{p}}$ is a weakly associated prime of $M_{\mathfrak{p}}$.
- iii. There exists an element $m \in M_{\mathfrak{p}}$ with $\sqrt{\text{Ann}_{R_{\mathfrak{p}}}(m)} = \mathfrak{p}R_{\mathfrak{p}}$

Proof.

$i. \implies ii.$

Assume $\mathfrak{p}$ is a weakly associated prime of M . Then there exists a $m \in M$ s.t. $\mathfrak{p}$ is minimal over $\text{Ann}_R(m)$.

Consider $\text{Ann}_R(m)R_{\mathfrak{p}} = \text{Ann}_{R_{\mathfrak{p}}}(m/1)$ this is true since $r/t \in \text{Ann}_{R_{\mathfrak{p}}}(m/1) \iff \exists u \in S : urm = 0 \in M \iff ur/ut \in \text{Ann}_R(m)R_{\mathfrak{p}} \iff r/t \in \text{Ann}_R(m)R_{\mathfrak{p}}$.

Also, since no prime $\mathfrak{q} \not\subseteq \mathfrak{p}$ contains $\text{Ann}_R(m)$ this implies that $\mathfrak{p}R_{\mathfrak{p}}$ is a minimal prime

over $\text{Ann}_R(m)R_p = \text{Ann}_{R_p}(m/1)$.

This is because if $I \subset R$ is an ideal then the minimal primes over IR_p are exactly qR_p where q is minimal over I in R with $q \subset p$.

Therefore, pR_p is a weakly associated prime of M_p .

ii. $\implies i$.

Assume pR_p is a weakly associated prime of M_p . Then there exists a $m/s \in M_p$ s.t. pR_p is minimal over $\text{Ann}_{R_p}(m/s) = \text{Ann}_{R_p}(m/1) = \text{Ann}_R(m)R_p$.

Let q be a prime of R minimal over $\text{Ann}_R(m)$ with $q \subset p$. Then qR_p is minimal over $\text{Ann}_R(m)R_p$, but this implies that $qR_p = pR_p$, therefore $q = p$. Therefore, p is minimal over $\text{Ann}_R(m)$ thus, p is a weakly associated prime of M .

ii. $\implies iii$.

Assume pR_p is a weakly associated prime of M_p . This implies there exists a $m/1 \in M_p$ s.t. pR_p is minimal over $\text{Ann}_{R_p}(m/1) = \text{Ann}_R(m)R_p$.

Since pR_p is maximal this implies that no prime $q \subsetneq pR_p$ contains $\text{Ann}_R(m)R_p$ by minimality. Since every prime is contained in pR_p , the only prime containing $\text{Ann}_R(m)R_p$ is pR_p .

Therefore, since $\sqrt{\text{Ann}_R(m)R_p}$ is the intersection of all primes that contain $\text{Ann}_R(m)R_p$ this implies that this is just equal to pR_p .

iii. $\implies ii$.

Assume there exists an element $m \in M_p$ s.t. $\sqrt{\text{Ann}_{R_p}(m)} = pR_p$.

Let q be any prime of R_p s.t. $\text{Ann}_{R_p}(m) \subset q \subset pR_p \implies \sqrt{\text{Ann}_{R_p}(m)} \subset q \subset pR_p \implies q = pR_p$. Therefore, pR_p is the unique minimal prime over $\text{Ann}_{R_p}(m)$ therefore, pR_p is a weakly associated prime of M_p . ■

(c)

Proof.

Let p be an associated prime then $\text{Ann}_R(m) = p$ for $m \in M$ then p is trivially minimal over $\text{Ann}_R(m)$. Hence, associated primes are a subset of weakly associated primes.

Let p be a weakly associated prime of M .

From problem set 5 since R is noetherian we have

$$q \in \text{Ass}_R(M) \iff qR_q \in \text{Ass}_{R_q}(M_q)$$

Therefore it suffices to show that $pR_p \in \text{Ass}_{R_p}(M_p)$

By (b) this implies that there exists a $m \in M_p$ s.t. $\sqrt{\text{Ann}_{R_p}(m)} = pR_p$.

Consider $R_p m \cong R_p / \text{Ann}_{R_p}(m)$. Since R is noetherian this implies R_p is also noetherian

therefore $\text{Ass}_{R_p}(R_p/\text{Ann}_{R_p}(m))$ is equal to the primes minimal over $\text{Ann}_{R_p}(m)$.

Since the only minimal prime over $\text{Ann}_{R_p}(m)$ is $\mathfrak{p}R_p$ this implies $\mathfrak{p}R_p \in \text{Ass}_{R_p}(R_p/\text{Ann}_{R_p}(m))$.

Therefore, there exists an $a \in R_p$ s.t. $\bar{a} \in R_p/\text{Ann}_{R_p}(m)$ has $\text{Ann}_{R_p}(\bar{a}) = \mathfrak{p}R_p$.

Then pulling back to $R_p m$, the element $am \in M_p$ satisfies

$$\text{Ann}_{R_p}(am) = \mathfrak{p}R_p$$

This is because $b \in \text{Ann}_{R_p}(am) \iff bam = 0 \iff ba \in \text{Ann}_{R_p}(m) \iff \bar{b}\bar{a} = 0 \in R_p/\text{Ann}_{R_p}(m) \iff \bar{b} \in \text{Ann}_{R_p}(\bar{a}) = \mathfrak{p}R_p$.

Therefore, $\mathfrak{p}R_p \in \text{Ass}_{R_p}(M_p)$, thus $\mathfrak{p} \in \text{Ass}_R(M)$.

Therefore, $\text{Ass}_R(M) = \{\mathfrak{p} \in \text{Spec}(R) \mid \mathfrak{p} \text{ is a weakly associated prime of } M\} = \text{wAss}_R(M)$

Then by (a) we get that

$$\text{wAss}_R(M) = \{\mathfrak{p} \in \text{Spec}(R) \mid \exists m \in M \text{ s.t. } V(\mathfrak{p}) \text{ is an irreducible component of } \text{Supp}_R(m)\} \quad (3\text{rd equality})$$

If $\mathfrak{p} \in \text{Ass}_R(M)$ then this implies $\text{Ann}_R(m) = \mathfrak{p} \implies V(\text{Ann}_R(m)) = \text{Supp}_R(m)$.

Thus, $\text{Supp}_R(m) = V(\mathfrak{p})$ is irreducible with generic point $\mathfrak{p}$ due to the bijection between irreducible closed subsets and prime ideals.

If $\text{Supp}_R(m)$ is irreducible with generic point $\mathfrak{p}$, then this implies that $\text{Supp}_R(m) = V(\mathfrak{p}) = V(\text{Ann}_R(m))$. Applying I this implies that $I(\text{Supp}_R(m)) = \mathfrak{p} = \sqrt{\text{Ann}_R(m)}$.

This implies that $\mathfrak{p}$ is minimal over $\text{Ann}_R(m)$, so $\mathfrak{p}$ is a weakly associated prime thus it is an associated prime.

(4th equality)

If $\mathfrak{p}$ is an associated prime then this implies that $\mathfrak{p} = \text{Ann}_R(m)$ for $m \in M$. Thus, $\text{Supp}_R(m) = V(\mathfrak{p}) \implies I(\text{Supp}_R(m)) = \mathfrak{p}$.

If $\text{Supp}_R(m) = V(\text{Ann}_R(m))$ is irreducible then by the bijection between irreducible closed subsets and primes we get that $I(\text{Supp}_R(m))$ is a prime ideal. This implies that $I(\text{Supp}_R(m)) = \mathfrak{p}$ for some prime. This implies with the same argument as above since $I(\text{Supp}_R(m)) = \sqrt{\text{Ann}_R(m)} = \mathfrak{p}$ that $\mathfrak{p}$ is an associated prime.

Therefore, all equalities have been shown. ■

Problem 3:

Let $\mathfrak{q} = (x, y^2) \subset k[x, y]$. Prove that $\mathfrak{q}$ is primary but is not a power of a prime ideal.

Proof.

Consider $\sqrt{\mathfrak{q}}$. First $x \in \mathfrak{q}$, so $x \in \sqrt{\mathfrak{q}}$. Also, $y^2 \in \mathfrak{q} \implies y \in \sqrt{\mathfrak{q}}$. Thus, $(x, y) \subset \sqrt{\mathfrak{q}}$, but since $\mathfrak{q} \subset (x, y)$ and (x, y) is maximal this implies $\sqrt{\mathfrak{q}} \subseteq (x, y)$.

Therefore, $\sqrt{\mathfrak{q}} = (x, y)$. Shown in class this implies that $\mathfrak{q}$ is (x, y) -primary since (x, y) is maximal.

If $\mathfrak{q} = \mathfrak{p}^n$ for some prime $\mathfrak{p}$, then $\sqrt{\mathfrak{q}} = \mathfrak{p} = (x, y)$. Reducing modulo (y) we get

$\mathfrak{q} = (x, y^2) \rightarrow (x)$, and $(x, y)^n \rightarrow (x^n)$.

If $\mathfrak{q} = (x, y)^n$, we would get $(x) = (x^n)$ in $k[x]$, which implies $n = 1$ which would imply $\mathfrak{q} = (x, y)$ which is a contradiction. Therefore, $\mathfrak{q}$ is not a power of a prime. ■

Problem 4:

Let $R = k[x, y, z]/(xy - z^2)$ where k is a field.

(a)

Show that the ideal $\mathfrak{p} = (\bar{x}, \bar{z})$ is prime.

Proof.

$R/\mathfrak{p} \cong k[x, y, z]/(xy - z^2, x, z) \cong k[y]$ which is a domain so $\mathfrak{p}$ is prime. ■

(b)

Prove that $\sqrt{\mathfrak{p}^2} = \mathfrak{p}$, but $\mathfrak{p}^2$ is not a primary ideal. Thus powers of prime ideals need not be primary in general.

Proof.

In any ring $\sqrt{I^n} = \sqrt{I}$, so $\sqrt{\mathfrak{p}^2} = \mathfrak{p}$.

Consider $\mathfrak{p}^2 = (x^2, xz, z^2) = (x^2, xz, xy)$

Consider $xy = z^2 \in \mathfrak{p}^2$, but $x \notin \mathfrak{p}^2$ and $y \notin \mathfrak{p}$. Thus, $\mathfrak{p}^2$ is not primary. ■

(c)

On the other hand, prove if A is a ring with a maximal ideal $\mathfrak{m}$, then for any integer $n > 0$ the ideal $\mathfrak{m}^n \subset A$ is primary.

Proof.

Suppose $ab \in \mathfrak{m}^n$ and $b \notin \mathfrak{m}$. Since $(\mathfrak{m}, b) = A \implies s + tb = 1$ for $s \in \mathfrak{m}$ and $t \in A$.

Then $1 = (s + tb)^n \in \mathfrak{m}^n + (b)$.

This implies there exists a $u \in A$ and $r \in \mathfrak{m}^n$ with $ub + r = 1$. Thus, $a = (au)b + ar \in \mathfrak{m}^n$.

Therefore, $a \in \mathfrak{m}^n$. Thus, $\mathfrak{m}^n$ is $\mathfrak{m}$ -primary. ■

(d)

Going back to the ring R , find a minimal primary decomposition for the ideal $\mathfrak{p}^2 \subset R$.

Proof.

WTS: $\mathfrak{p}^2 = (x) \cap (x, y, z)^2 = (x) \cap (x^2, xy, xz, y^2, yz, z^2)$.

First, $\mathfrak{p}^2 = (x^2, xy, xz) \subset (x)$ and $\mathfrak{p}^2 \subset (x, y, z)^2$

If $f \in (x) \cap (x, y, z)^2$ then $f = xg$ for $g \in R$. If g had a nonzero constant term, f

would have a degree-1 component this contradicts $f \in (x, y, z)^2$ since $(x, y, z)^2$ has no polynomials with any degree one components. Therefore, $g \in (x, y, z)$, since it has no constant term.

Therefore, $f \in \mathfrak{p}^2 = x(x, y, z)$.

Thus, $\mathfrak{p}^2 = (x) \cap (x, y, z)^2$ which is a minimal primary decomposition since (x) is prime so it is (x) -primary and $(x, y, z)^2$ is (x, y, z) -primary since it is a power of a maximal ideal.

Also, it is irredundant. ■

(e)

Find the associated primes of the R -module $R/\mathfrak{p}^2$.

Proof.

Since $k[x, y]$ is noetherian this implies R is noetherian, and shown in class the associated primes of R/I are exactly the radicals of the primary components in a minimal primary decomposition of I .

Thus, $\text{Ass}_R(R/\mathfrak{p}^2) = \{\mathfrak{p}, (x, y, z)\}$ ■

Problem 5:

Let $I \subset R$ be an ideal with primary decomposition $I = q_1 \cap \cdots \cap q_n$. Let $\mathfrak{p}_i = \sqrt{q_i}$. Prove that the minimal elements in the set $\{\mathfrak{p} \in \text{Spec}(R) \mid I \subset \mathfrak{p}\}$ coincide with the minimal elements in the set $\{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$

Proof.

Let $\mathfrak{p}$ be a minimal prime over I . Then $I \subset \mathfrak{p} \implies \sqrt{I} \subset \mathfrak{p}$. Since $\sqrt{I} = \sqrt{\bigcap q_i} = \bigcap \sqrt{q_i} = \bigcap \mathfrak{p}_i$.

Since $\prod \mathfrak{p}_i \subset \bigcap \mathfrak{p}_i$ we have $\sqrt{\prod \mathfrak{p}_i} \subset \bigcap \mathfrak{p}_i$.

Let $x \in \bigcap \mathfrak{p}_i$ then $x \in \mathfrak{p}_i$ for all i . This implies that $x^n \in \prod \mathfrak{p}_i$ thus $x \in \sqrt{\prod \mathfrak{p}_i}$.

Therefore, $\sqrt{\prod \mathfrak{p}_i} = \bigcap \mathfrak{p}_i$.

This implies that $\prod \mathfrak{p}_i \subset \mathfrak{p}$ since $\mathfrak{p}$ is radical. Since $\mathfrak{p}$ is prime if none of the $\mathfrak{p}_i$ were contained in $\mathfrak{p}$ we could pick $a_i \in \mathfrak{p}_i \setminus \mathfrak{p}$ for each i . Then

$a_1 a_2 \dots a_n \in \prod \mathfrak{p}_i \subset \mathfrak{p}$. Since $\mathfrak{p}$ is prime and $a_1 \dots a_n \in \mathfrak{p}$ it follows that some $a_j \in \mathfrak{p}$ contradicting the choice $a_j \notin \mathfrak{p}$.

Thus, there must be some j with $\mathfrak{p}_j \subset \mathfrak{p}$, but each $\mathfrak{p}_j$ contains I , so by minimality of $\mathfrak{p}$, we have $\mathfrak{p} = \mathfrak{p}_j$. Moreover, if some $\mathfrak{p}_k \subset \mathfrak{p}_j$, then $\mathfrak{p}_k$ would also contain I , contradicting the minimality of $\mathfrak{p}$. Thus, $\mathfrak{p}$ is minimal among $\{\mathfrak{p}_i\}$.

Let $\mathfrak{p}_j$ be minimal among $\{\mathfrak{p}_i\}$. Suppose there is a prime $\mathfrak{q}$ with $I \subset \mathfrak{q} \subsetneq \mathfrak{p}_j$.

Then, as above, $\prod \mathfrak{p}_i \subset \mathfrak{q}$, so $\mathfrak{p}_k \subset \mathfrak{q}$ for some k . But then $\mathfrak{p}_k \subset \mathfrak{q} \subsetneq \mathfrak{p}_j$, which contradicts the minimality of $\mathfrak{p}_j$ among the $\mathfrak{p}_i$. Therefore, $\mathfrak{q}$ does not exist and $\mathfrak{p}_j$ is minimal over I .

Therefore, the minimal primes over I are exactly the minimal elements among $\{\mathfrak{p}_i\}$. ■

Problem 6:

Let $\mathbb{C}[x_1, x_2, \dots]$ be the polynomial ring on infinitely many generators $x_i, i \in \mathbb{Z}_{>0}$ over the complex numbers, let $(x_1x_2, x_3x_4, x_5x_6, \dots)$ be the ideal generated by the products $x_i x_{i+1}$ for odd $i \in \mathbb{Z}_{>0}$, and let $R = \mathbb{C}[x_1, x_2, \dots] / (x_1x_2, x_3x_4, x_5x_6, \dots)$.

(a)

Prove that R contains infinitely many minimal prime ideals.

Proof.

For any subset $T \subset \mathbb{Z}_{>0}$, choose exactly one variable from each pair $\{x_{2k-1}, x_{2k}\}$. take x_{2k-1} if $k \in T$ and x_{2k} if $k \notin T$.

Define $\mathfrak{p}_T = (\{\bar{x}_{2k-1} \mid k \in T\} \cup \{\bar{x}_{2k} \mid k \notin T\})$.

Let $K_T = \{x_{2k-1} \mid k \in T\} \cup \{x_{2k} \mid k \notin T\}$

Consider $R/\mathfrak{p}_T \cong \mathbb{C}[x_1, x_2, \dots] / (I + (K_T))$

For each k , the generator $x_{2k-1}x_{2k} \in I$ has at least one factor in (K_T) .

This implies that $I \subset (K_T)$, so $I + (K_T) = (K_T)$.

Therefore, $R/\mathfrak{p}_T \cong \mathbb{C}[x_1, x_2, \dots] / (K_T) \cong \mathbb{C}[\{x_1, x_2, \dots\} \setminus K_T]$ which is an integral domain.

Therefore, $\mathfrak{p}_T$ is prime.

Consider $\mathfrak{p}_T$ for some subset $T \subset \mathbb{Z}_{>0}$.

Let $\mathfrak{p} \subset \mathfrak{p}_T$.

For each k , we have $\bar{x}_{2k-1}\bar{x}_{2k} = 0 \in \mathfrak{p}$, so at least one factor is in $\mathfrak{p}$ for all k .

However, for each k , if $k \notin T$ then $\bar{x}_{2k-1} \notin \mathfrak{p}_T \implies \bar{x}_{2k-1} \notin \mathfrak{p} \implies \bar{x}_{2k} \in \mathfrak{p}$.

Also, if $k \in T$ then $\bar{x}_{2k} \notin \mathfrak{p}_T \implies \bar{x}_{2k} \notin \mathfrak{p} \implies \bar{x}_{2k-1} \in \mathfrak{p}$. Thus, $\mathfrak{p}$ contains all generators of $\mathfrak{p}_T$. Thus, $\mathfrak{p}_T = \mathfrak{p}$.

Therefore, $\mathfrak{p}_T$ is minimal. Thus, R has infinitely many minimal primes. ■

(b)

Prove that the zero ideal $(0) \subset R$ does not have a primary decomposition.

Proof.

If (0) has a primary decomposition, it would be a finite intersection, so R would only have finitely many primes over (0) from problem 5. This is a contradiction since R has infinitely many minimal primes which means infinitely many over (0) . Thus, (0) does not have a primary decomposition in R . ■

Problem 7:

Let R be a ring and $S \subset R$ be a multiplicative system.

(a)

Let $\mathfrak{p} \subset R$ be a prime ideal and let $\mathfrak{q} \subset R$ be a $\mathfrak{p}$ -primary ideal. Prove the following:

- i. $S^{-1}\mathfrak{q} = S^{-1}R$ if and only if $\mathfrak{p} \cap S \neq \emptyset$.
- ii. If $\mathfrak{p} \cap S = \emptyset$, then extension and contraction of ideals along the ring homomorphism $R \rightarrow S^{-1}R$ provides a bijection between $\mathfrak{p}$ -primary ideals $\mathfrak{q} \subset R$ and $S^{-1}\mathfrak{p}$ -primary ideals in $S^{-1}R$.

Proof.

Let $\mathfrak{p} \subset R$ be prime and $\mathfrak{q} \subset R$ be $\mathfrak{p}$ -primary.

i.:

Assume $S^{-1}\mathfrak{q} = S^{-1}R$. This implies $\mathfrak{q} \cap S \neq \emptyset$. Thus, let $s \in \mathfrak{q} \cap S$. Thus, $s \in \mathfrak{q} \implies s \in \sqrt{\mathfrak{q}} = \mathfrak{p}$. Therefore, $\mathfrak{p} \cap S \neq \emptyset$.

Assume $\mathfrak{p} \cap S \neq \emptyset$. Let $s \in \mathfrak{p} \cap S$. Then this implies $s \in \mathfrak{p} = \sqrt{\mathfrak{q}} \implies s^n \in \mathfrak{q}$ for $n \in \mathbb{N}$, but since S is a multiplicative system this implies $s^n \in S$. Therefore, $\mathfrak{q} \cap S \neq \emptyset$.

ii.:

Assume $\mathfrak{p} \cap S = \emptyset$.

If $\mathfrak{q}$ is $\mathfrak{p}$ -primary then

Since $\sqrt{\mathfrak{q}} = \mathfrak{p}$ and $\mathfrak{p} \cap S = \emptyset$, this implies that $\sqrt{S^{-1}\mathfrak{q}} = S^{-1}\sqrt{\mathfrak{q}} = S^{-1}\mathfrak{p}$.

Let $a/s \cdot b/t \in S^{-1}\mathfrak{q}$ and $a/s \notin S^{-1}\mathfrak{q}$, then there is a $u \in S$ s.t. $uab \in \mathfrak{q}$.

If $b/t \notin S^{-1}\mathfrak{p}$ then $b \notin \mathfrak{p}$. Since $\mathfrak{q}$ is $\mathfrak{p}$ -primary and $a(ub) \in \mathfrak{q}$ with $ub \notin \mathfrak{p}$, we get $a \in \mathfrak{q} \implies a/s \in S^{-1}\mathfrak{q}$ which is a contradiction. Thus, $b/t \in S^{-1}\mathfrak{p}$. Therefore, $S^{-1}\mathfrak{q}$ is $S^{-1}\mathfrak{p}$ -primary.

If Q is $S^{-1}\mathfrak{p}$ -primary, then let $Q^c = \mathfrak{q}$.

$\sqrt{\mathfrak{q}} = (\sqrt{Q})^c = (S^{-1}\mathfrak{p})^c = \mathfrak{p}$. If $ab \in \mathfrak{q}$ and $b \notin \mathfrak{p}$, then $a/1, b/1 \in Q$ and $b/1 \notin S^{-1}\mathfrak{p}$, so $a/1 \in Q \implies a \in \mathfrak{q} = Q^c$. Thus, $\mathfrak{q}$ is $\mathfrak{p}$ -primary.

Therefore, this is a bijection. ■

(b)

Let $I \subset R$ be an ideal with a primary decomposition $I = q_1 \cap \cdots \cap q_n$, and let $p_i = \sqrt{q_i}$. Prove that $S^{-1}I$ is a proper ideal in $S^{-1}R$ if and only if there exists an i s.t. $S \cap p_i = \emptyset$, and in this case prove that

$$S^{-1}I = \bigcap_{p_i \cap S = \emptyset} S^{-1}q_i \text{ and } \varphi^{-1}(S^{-1}I) = \bigcap_{p_i \cap S = \emptyset} q_i$$

are primary decompositions, where $\varphi : R \rightarrow S^{-1}R$ is the localization homomorphism.

Proof.

Let $I = \bigcap q_i$ be a primary decomposition.

First $S^{-1}I = \bigcap_i S^{-1}q_i$. By (a), $S^{-1}q_i = S^{-1}R$ iff $p_i \cap S \neq \emptyset$. Therefore,

$S^{-1}I$ is proper $\iff \exists i$ with $S^{-1}q_i \neq S^{-1}R \iff \exists i$ with $p_i \cap S = \emptyset$

In the proper case we have,

$$S^{-1}I = \bigcap S^{-1}q_i = \bigcap_{p_i \cap S = \emptyset} S^{-1}q_i$$

By (a) we also know that each $S^{-1}q_i$ is $S^{-1}p_i$ -primary, thus $\bigcap_{p_i \cap S = \emptyset} S^{-1}q_i$ is a primary decomposition in $S^{-1}R$.

Again by (a) using the bijection we get that

$$\varphi^{-1}(S^{-1}I) = \varphi^{-1}\left(\bigcap_{p_i \cap S = \emptyset} S^{-1}q_i\right) = \bigcap_{p_i \cap S = \emptyset} \varphi^{-1}(S^{-1}q_i) = \bigcap_{p_i \cap S = \emptyset} q_i$$

(c)**Proof.**

Let $I = \bigcap q_i$ be a minimal primary decomposition. Fix i s.t. p_i is minimal among $\{p_j\}$.

Let $S = R \setminus p_i$

By (b) applied to S , only the components with $p_j \cap S = \emptyset$ survive upon localizing at p_i , but since p_i is minimal in the set $\{p_j\}$, the only such j is $j = i$.

Therefore, $IR_{p_i} = q_i R_{p_i}$.

Now contradicting back we get $\varphi^{-1}(IR_{p_i}) = \varphi^{-1}(q_i R_{p_i}) = q_i$, where the last equality uses (a).